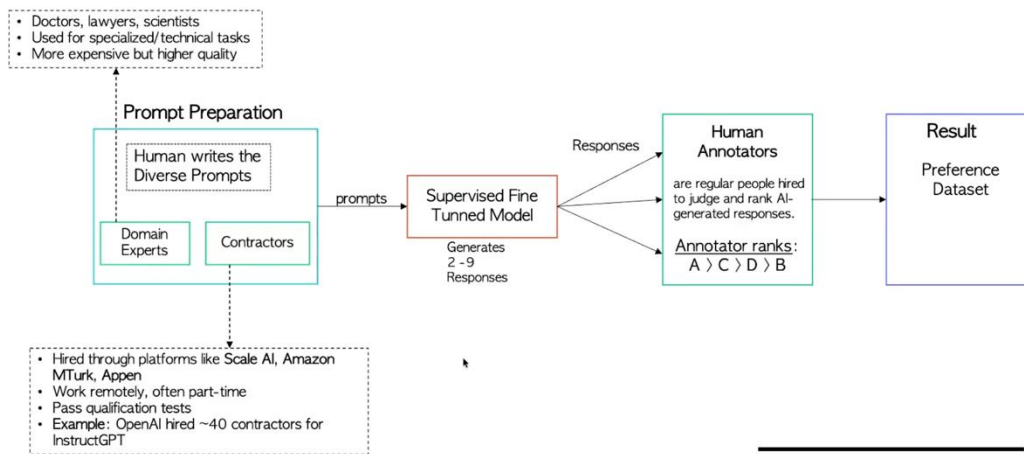


Fine Tuning – Time for Human Evaluation(Stage-1).

Approach 1



In this we are going to hire all the different people from diff domain skills like lawyers, doctors, engineers, and many more and they are going to ask to develop the prompts ...

Now we are going to collect all these prompts and then going to pass it on to our Model (Supervised Fine Tuned Model) which in deed analyses them and genrate 2-9 responses ..

Now these responses will be further evaluated by humans (who are special hires) then they will provide the final headsup and then provide the ranking for them accordingly ..

	human_preference	score_a	score_b	score_c	annotator_notes
prints 0,1,2,3,4.	A > C > B	4	-3	2	Response: A is comprehensive and clear. C is concise but lacks detail.
Cook in pan 4) Stir 5	A > C > B	5	-2	1	A provides perfect detail. C is too brief but correct. B doesn't answer
	B > A > C	-1	5	-2	B is perfect for a child. A is too technical. C is unhelpfully vague.
mon OTC options exi:	C > A > B	2	-4	4	C appropriately disclaims medical advice. A is helpful but could be se
untries. It was a very	A > B > C	5	1	-3	A provides comprehensive answer. B is correct but minimal. C doesn't
ompany, prepare you	A > C > B	5	-2	2	A is empathetic and helpful. C is practical but less warm. B is dismiss
a decimal and multij	A > B > C	5	3	-2	A is concise with answer and method. B is slightly wordy. C doesn't g
e by adding print stat	A > C > B	5	-1	1	A explains the error with example. C gives generic advice. B is too bri
	A > C > B	5	-2	0	A gives comprehensive list. C mentions only one place. B is too gener
	A > C > B	5	-3	-1	A presents nuanced view. C is too absolutist. B promotes problemat
	A > B > C	5	-1	-2	A provides reasoned analysis. B is unhelpfully vague. C is dismissive a
vocabulary first befo	A > B > C	5	0	-2	A gives actionable strategies. B is correct but vague. C is discouraging
cycles and human ac	A > C > B	5	-3	1	A provides accurate scientific consensus. C is vague but not wrong. B
bies, personal interes	A > B > C	5	1	-3	A is comprehensive and strategic. B is minimal. C includes bad advice
w recipes exactly or i	A > B > C	5	0	-3	A provides detailed guidance with ratios. B is too brief. C is incorrect
ves in the yard.	A > B > C	5	1	-2	A provides nuanced, helpful advice. B is minimal. C gives poor advice
ound interest.	A > B > C	5	-1	-1	A provides balanced, nuanced advice. B and C are too absolutist.
	A > C > B	5	-2	-1	A lists multiple possibilities and solutions. C oversimplifies. B is alarm
	A > C > B	5	-2	0	A is encouraging and specific for beginners. C is vague. B is unhelpful
	A > C > B	5	-1	-2	A provides systematic troubleshooting. B jumps to conclusion. C is us

People whom we hired they will evaluate the entire responses and provide the score.

Fine Tuning – Human Annotators

Are they AI experts? No! They're regular people (often college-educated) who are trained to evaluate quality.

How do they know what's "good"? They follow detailed guidelines that define quality criteria (helpful, accurate, safe, clear).

Q: Can they be biased? Yes! That's why companies use:

- Diverse annotator pools
- Multiple annotators per task
- Clear objective rubrics

Q: Why not just use AI to rank? Humans still better at:

- Nuanced quality judgment
- Understanding context and culture
- Detecting subtle harmful content
- Knowing what's truly helpful

Payment is PER TASK (1 Task = 1 Prompt + Ranking ALL the Responses)

Task Complexity	Pay per Task	Time Needed	Effective Hourly Rate
Simple (rank 2-3 responses)	\$0.50 – \$2	1-2 min	\$30-40/hour
Standard (rank 4-5 responses)	\$1.50 – \$5	3-5 min	\$40-50/hour
Complex (rank 6-9 + explanation)	\$3 – \$10	6-10 min	\$40-60/hour
Expert (medical, legal, etc.)	\$20 – \$100	15-30 min	\$80-120/hour

Project costs:

- Small dataset (10K comparisons): \$5K – \$20K
- Medium (50K comparisons): \$50K – \$150K
- Large scale (100K+): \$200K – \$1M+

Model training is not an easy task its super expensive. Here we are going to hire multiple people not required to be the experts but good to do what is asked ..

As usual nothing comes for free, these people have to paid for the work what they do. You can check the prices as per the screen shot above.

Fine Tuning – Time for Human Evaluation (Stage- 1).

Approach 2

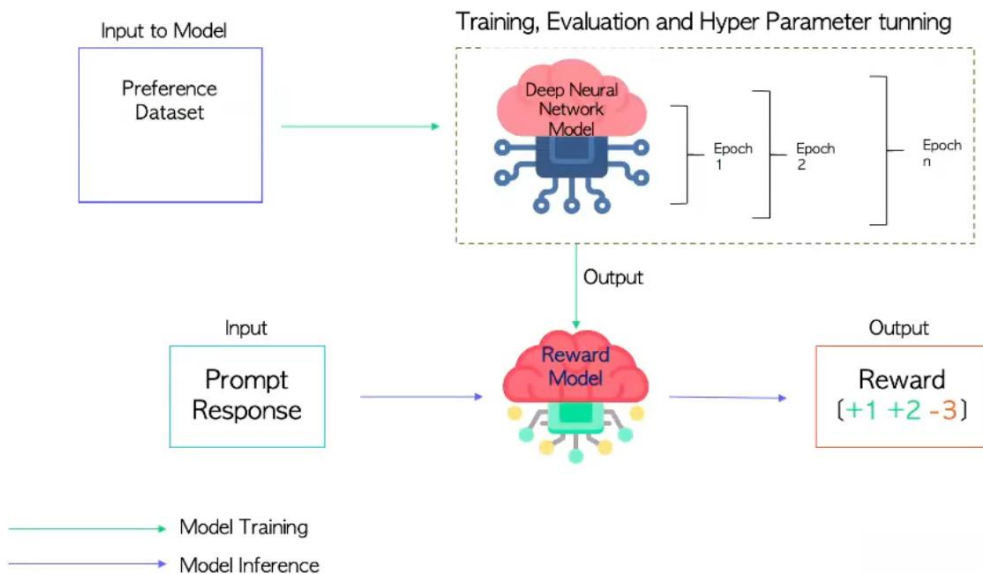


In the approach2 we just bring up the real users and ask them to provide the prompts where we generate and learn bad prompts vs good prompts and so on ...

Our super fine tuned model then finally be ready with such huge data sets then it will start producing the responses. Now this data we call it as "preference dataset" which we can use it on further models where we should able to generate good results.

Now since our model got trained with all the inputs how a human is going to think and react same way our model which we are calling it as "reward model" is going to replace the humans.

Time to Replace Human with Model – Reward Model



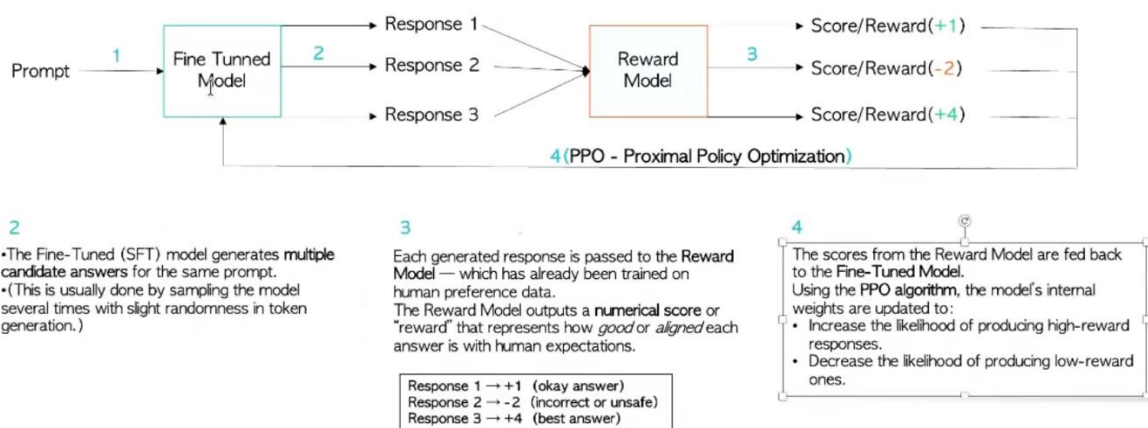
In this phase we are going to provide all our preference datasets to the Neural Network model which in deed decides and generate the rewards. Earlier these rewards are generated by humans but now its been generated by your reward model.

Reinforcement Learning

After Supervised Fine-Tuning (SFT), the model can follow instructions, but it doesn't yet *understand which answers are better*.

Reinforcement Learning fixes that by teaching the model through **feedback and scoring** — i.e., "rewarding good answers" and "penalizing bad ones."

This process is called Reinforcement Learning from Human Feedback (RLHF).



All the journey we had till now is to generate a fine tune model. Now we have to ensure that this model is learning the things properly and we have to confirm that what is good and bad for it ..

So in the phase of reinforcement what we are going to do is

Human prompt --> fine tuned model --> generate two or 3 response .. Now these responses will be send to the reward model. What exactly this reward model will do ?? This model helps in generating the rewards for all the responses that our fine tune model gave.

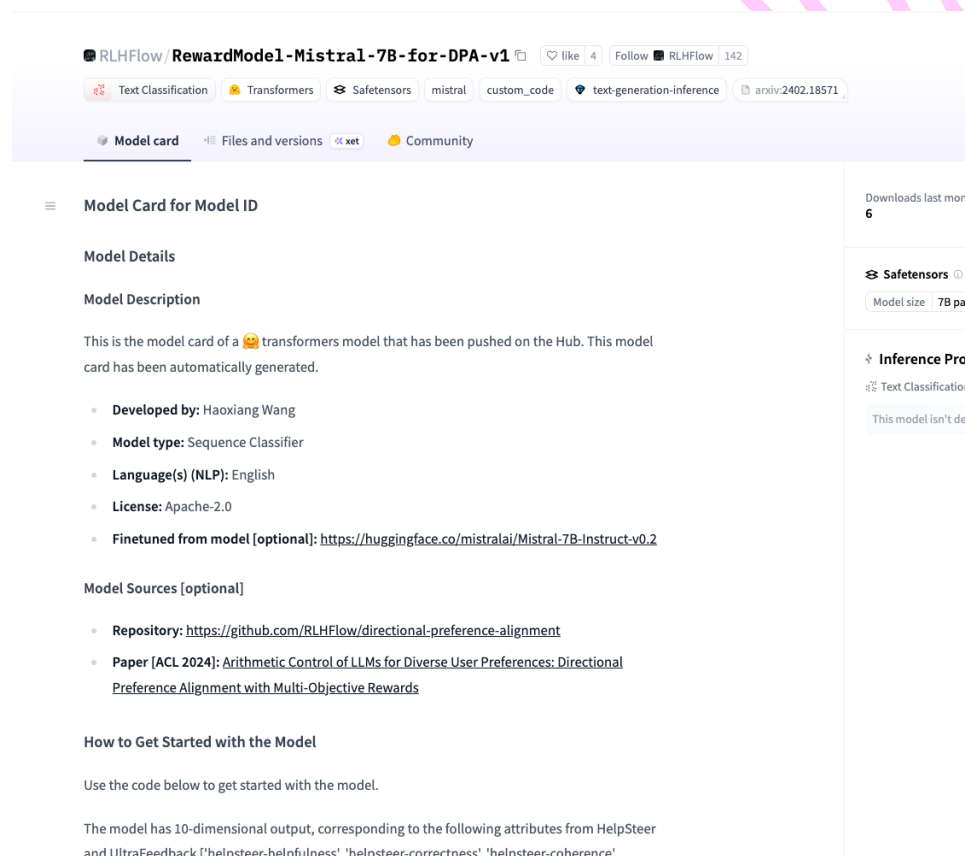
Here we have to teach the fine tune model about the pros and cons on its responses it generated.. In simple I have to tell the fine tune model that score for response 1 is good and for others are not that good.

So who will take this responsibility of teaching fine tune model? Here comes your **Proximity policy optimization** come to picture .. Which take the role of reinforcing fine tune model to learn best responses.

Hence this is a continuous cycle process where we are going to fine tuned model.

Hugging face is the platform which provides free/paid models, for example

<https://huggingface.co/RLHFlow/RewardModel-Mistral-7B-for-DPA-v1>



The screenshot displays the Hugging Face model card for **RLHFlow/RewardModel-Mistral-7B-for-DPA-v1**. The card includes the following sections:

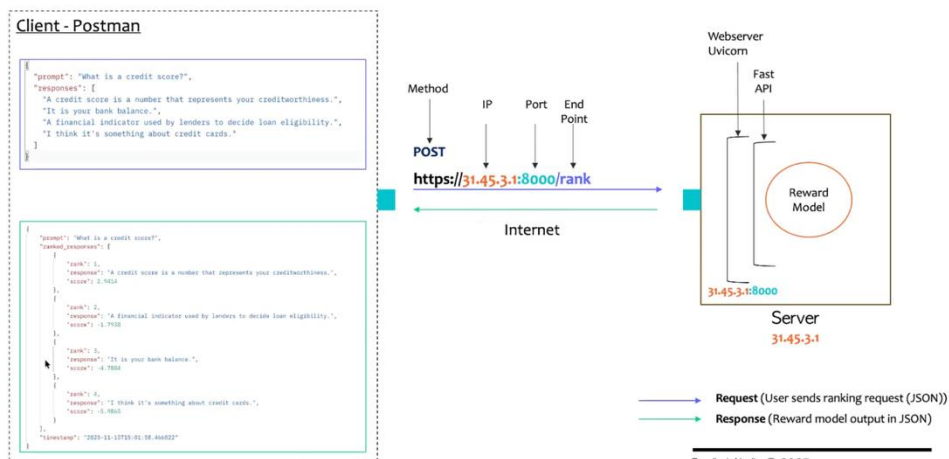
- Model Details:** Shows the model name, likes (4), and follows (142).
- Model Description:** States that this is a transformers model pushed on the Hub, with an automatically generated card.
- Model Sources [optional]:** Lists the repository (<https://github.com/RLHFlow/directional-preference-alignment>) and a paper from ACL 2024 titled "Arithmetic Control of LLMs for Diverse User Preferences: Directional Preference Alignment with Multi-Objective Rewards".
- How to Get Started with the Model:** Provides instructions on how to use the code and mentions the model's 10-dimensional output corresponding to attributes from HelpSteer and UltraFeedback.

On the right side of the card, there are additional details such as "Downloads last month: 6", "Safetensors" information, and "Inference Providers" (Text Classification).

Now we can go ahead and use this model to get the rewards for all the prompts users are passing.

We can integrate this api call and get the rewards for all the responses my finetune model generating

Let's look at Reward Model in Action



Now our reward models help us to generate the score for the prompts. Question is how are they going to provide the reward points ??

These reward points are based on "graders" like honesty, relevance, safety, helpfulness, format, code quality grader and so on ..

Read below article for more information on reward models:

<https://aws.amazon.com/blogs/machine-learning/revolutionize-customer-satisfaction-with-tailored-reward-models-for-your-business-on-amazon-sagemaker/>